

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 2.2 CHEMICAL FAMILIES – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle. Remember Mama Wanjiru, the food seller in Mombasa, who asked why her aluminium sufuria is safe while chlorine in her tap water is poisonous as a gas. Use the evidence you collected in Lessons 1 to 8, the class family model and the Driving Question Board (DQB) to answer each question in the space below it. Write in full sentences. For each answer, use a claim (what you think), evidence (data or observations from our lessons) and reasoning (the idea of outer electrons and position in the periodic table that links them). Use the words element, compound, group, period and chemical family correctly. Write balanced equations with $\rightarrow$ where asked. Use these data when you need them. Sodium (Na): density 0.97 g/cm³, melting point 98 C. Magnesium (Mg): density 1.74 g/cm³, melting point 650 C. Aluminium (Al): density 2.70 g/cm³, melting point 660 C. Copper (Cu): density 8.96 g/cm³, melting point 1085 C, excellent conductor. Lead (Pb): density 11.34 g/cm³, melting point 327 C. Water has a density of 1.00 g/cm³. Chlorine is added to Mombasa water at about 0.5 mg/L.

Part 1: Why is sodium dangerous alone?

In the demonstration, Mr. Otieno dropped a pea-sized piece of sodium into water behind a safety screen. (a) Describe what you saw and what the pink colour with phenolphthalein showed. (b) Which chemical family does sodium belong to, and how many outer electrons does each atom have? Write its electron arrangement. (c) Explain why sodium reacts so violently with water. (d) Write the balanced equation for the reaction. (e) Lithium, sodium and potassium are in the same group. Explain how their reactivity compares and why.

Write your answer here:

Part 2: Sodium, chlorine and safe salt, and chlorine in Mombasa water

Mama Wanjiru asks why chlorine in her tap water is safe when chlorine as a gas is poisonous, and why the sodium that is dangerous alone is in the salt she puts in her food. (a) Which chemical family is chlorine in, and how many outer electrons does it have? Write its electron arrangement. (b) Write the balanced equation for sodium reacting with chlorine. (c) Use the idea of outer electrons and what you know about the dose to explain to Mama Wanjiru why the tap water is safe. (d) Use outer electrons to explain why table salt is safe to eat even though sodium alone is dangerous.

Write your answer here:

Part 3: Reactivity and position in the periodic table

Akinyi asked, 'If sodium is so dangerous, are all metals like that?' (a) Describe what happened when you tested magnesium ribbon in cold water, in dilute HCl and in air. (b) Compare magnesium with sodium using their electron arrangements and the evidence. (c) Write the balanced equations for magnesium burning in air and for magnesium reacting with dilute HCl. (d) Neon and argon do not react at all. Using outer electrons, explain why, and state a use that depends on this.

Write your answer here:

Part 4: Matching metals to jobs

Use the data given in the instructions and what you learned about the metals in the puzzle. (a) Using the data, give reasons why aluminium is used for Mama Wanjiru's sufuria. (b) Why is copper used for the wire in a kettle or jiko cable? (c) Why is the mabati roof made of iron coated with zinc? (d) Lead is the densest metal in the data but its use is being reduced. Explain why, using what you learned. (e) Compare sodium with aluminium using the data and explain why sodium would not suit a sufuria.

Write your answer here:

Part 5: Answering the driving question

Driving question: Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life? (a) Write a final explanation of the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle using claim, evidence and reasoning. Include outer electrons and position in the periodic table. (b) A jua kali artisan in Gikomba wants to make a new product: a cooking pot, a cable or a roof sheet. Choose one product, recommend an element and justify your choice using properties, safety, cost and availability. (c) How has your thinking changed since your first model in Lesson 1?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of outer electrons and periodic table position	Correctly links group, outer electrons, electron arrangements and reactivity for Na, Mg, Cl and noble gases, including the trend down Group 1.	Gives correct outer electron numbers and links most to reactivity, with minor gaps in the reasoning.	Gives few outer electron facts or does not link them to behaviour or position.
Use of evidence from the lessons	Uses accurate observations and data (sodium demo, Mg tests, 0.5 mg/L, densities and melting points) to support every claim.	Uses some accurate evidence but misses some data or gives it loosely.	Gives little evidence, or the data is inaccurate.
Chemical equations and terms	All equations are balanced and correct; element, compound, group and chemical family are used correctly.	Most equations are correct with one small error; terms mostly correct.	Equations are unbalanced or missing; terms are confused.

Matching element to job and trade-offs	Explains each job by properties and gives a justified recommendation weighing safety, cost and availability, including lead's toxicity.	Matches properties to jobs but gives limited discussion of trade-offs.	States uses without linking them to properties.
Claim-evidence-reasoning and explaining the puzzle	Gives a clear claim, evidence and reasoning that fully explains why sodium is dangerous alone but safe as salt, and why chlorine is safe at a tiny dose.	Explains most of the puzzle, with some reasoning missing.	Gives a partial or unclear explanation of the puzzle.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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